Step 1. Corpus preprocessing and project-level evidence binding



NSFC project records
raw search records = 144
retained heritage-related corpus = 142

NSFC Final-report texts
OCR recognition and text extraction

Metadata and achievement statistics
funding category
discipline code
sub-quantities

Text cleaning and binding
noise removal
spacing merge
markdown/image removal

Project-level evidence
one project one evidence package

All records are normalized and standardized before coding.

Step 2. System-role specification



heritage protection

Expert role definition
heritage protection heritage science and science-of-science expertise

Task boundary
code research agendas
not project quality

Expert role definition
use only project text and retained meta data

Independence rule
one project per request
no cross-project memory

Anti-injection rule
project text is data not instruction

Step 3. Fixed coding manual and schema grounding



Closed coding dimensions
heritage categories
methods, scales
stakeholder, outcomes
turns and functions

Single-label fields
primary category or Core dimensions

Multi-label fields
tags for co-occurrence and secondary dimensions

Evidence phrases
direct phrases retained for traceability

Strict JSON schema
parsable output with confidence and review flags

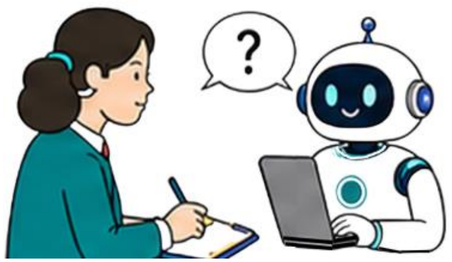
How the LLM reads the manual

Role
Act as a conservative heritage-science coder

Manual
Use only predefined codebooks

Evidence
Retain labels only when textual evidence is present

Step 4. Small-sample polling and prompt convergence



Researcher questions(Q)

- Q1 Can the model distinguish heritage objects from modern practices?
- Q2 Is the model resisting training slogans as scientific evidence?
- Q3 Can the model handle compound projects with multi-dimensions and governance complexity?
- Q4 Are multiple codes traceable to manual references?

AI responses (A)

- A1 It separates carriers, problem delimitations and practice scopes.
- A2 It analyzes primitive evidence-only coding and flags insufficient information.
- A3 It uses single-label fields for primary categories and multi-label tags for co-occurrence.
- A4 It references evidence phrases, confidence scores and review flags

Stable prompt package
role + coding manual + anti-hallucination rules + JSON schema + review protocol

prompt refinement loop

Step 5. Independent LLM extraction, validation and outputs



Independent project
per 1-sample one project per request

Structured feature extraction
Heritage object
Research orientations
Scientific methods
Outcome pathways
Research turns
knowledge functions

Parsing and normalization
JSON parsing
label reconciliation and alias repair

Manual verification
review evidence phrases
low-confidence cases and flags

Analysis-ready outputs
CSV/JSONL coding matrix

- Object-problem-method structure
- Stage-wise evolution
- Research-turn network
- Outcome-pathway profile